Brett Meggison

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Technical Skills

Fortran, Mathematica, FORM, Linux/bash script, gnuplot, SLURM like job managers, parallel computing, quantum computing, TeX, mathematical modeling (statistics, tensor algebra/calculus, group theory), python, tableau, and C

Experience

Brandon University/University of Manitoba/University of Winnipeg Brandon/Winnipeg, MB
Research - Computational Modeling Sept. 2016 Dec. 2026

- Wrote technical documents clearly communicating complicated information for intended audience.
- Iteratively built up complex mathematical models and implemented them through custom software.
- Developed and tested tens of thousands of lines of independently written code, across many languages.
- Organized and interpreted large volumes of technical data, then presented insights to a variety of audiences from highly technical fellow researchers, to less experienced under-graduates.
- Taught/assisted other researchers and students in group to use different technical tools/coding languages.
- Recognized informative contributions and areas of improvement to be made in existing literature.
- Integrated existing research and findings into own work, making it more effective and efficient.

Brandon University Brandon, MB
Sessional Teaching Sept. 2022 Apr. 2023

- Taught introductory physics - communicated technical concepts clearly to a diverse, non-technical audience.
- Designed and delivered lectures, assignments, and tests for the class.
- Coordinated with lab instructors to deliver lectures which were cohesive with lab material.
- Received very positive class evaluations.

Education

University of Manitoba Winnipeg, MB

PhD, Physics Graduation Date: Dec. 2026 Thesis: 3-Loop, 3PI Effective Action

Computationally solving systems of non-perturbative field theory equations, specifically using Schwinger-Dyson and n-particle irreducible (nPI) methods.

University of Manitoba Winnipeg, MB

MSc, Physics Graduation Date: Feb. 2020 Thesis: Anisotropic Graphene

Computationally solving systems of non-perturbative field theory equations, specifically using Schwinger-Dyson equations to evaluate the critical coupling in graphene.

Brandon University Brandon, MB

BSc (Hons), Physics, Minor in Mathematics Graduation Date: May 2016 Thesis: The 2PI effective theory at next-to-leading order using the functional renormalization group

Computationally solving systems of non-perturbative field theory equations, specifically using the 2PI effective action to test the effect of higher loop order on the pressure calculation of a high temperature system.

Google Advanced Data Analytics Certificate Oct. 2026
Methods for calculating, interpreting, and presenting actionable business insights from data sets. Common statistical and machine learning models used by data professionals in python.

Waskada Collegiate Institute Waskada, MB
Graduation Date: Jun. 2011

Interests: Gym, reading, movies, music, helping with family ranch